LiteGraphFuse: Distilling Fused Retrieval for Small Language Models

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Abstract

GraphFuse is accurate but slow, which limits its use with small language models and on modest hardware. LiteGraphFuse distils the GraphFuse fusion scorer into a 3-billion-parameter student and prunes the graph walk to two expansions per hop. On KaveriHop it reaches 55.0 and 43.9 EM, 95% of GraphFuse, at 105 ms per query instead of 420 ms. On the Tamil VaigaiQA benchmark it reaches 38.6 EM, above both Dense-Hop and GraphWalker-QA.

Keywords: distillation, efficiency, low-resource languages, hybrid retrieval

1 Introduction

GraphFuse [1] fused dense and graph retrieval, reaching 46.3 EM on 3-hop KaveriHop and 22.7 EM on 4-hop TriHop-Bench, at roughly twice the cost of a single retriever. Earlier work on Tamil [2] found graph retrieval alone unhelpful.

2 Method

LiteGraphFuse extends GraphFuse [1]: a GraphFuse teacher labels 200,000 candidate chains and a 3B student learns to reproduce the fusion scores. We cap the walk at two expansions per hop.

3 Experiments

We evaluate on KaveriHop [3] and VaigaiQA [2], and measure latency for GraphFuse and LiteGraphFuse on the same GPU. Dense-Hop [4] and GraphWalker-QA [5] numbers on VaigaiQA are quoted from the VaigaiQA paper. Distillation keeps 95% of GraphFuse's accuracy at about one quarter of its latency. Fusion helps on Tamil even though graph retrieval alone does not.

Method	Source	Kaveri-Hop 2-hop EM	Kaveri-Hop 3-hop EM	Kaveri-Hop latency (ms/query)	VaigaiQA 2-hop EM
GraphFuse	[1]	57.9	46.3	–	–
GraphFuse	ours	–	–	420	–
LiteGraphFuse	ours	55.0	43.9	105	38.6
Dense-Hop	[2]	–	–	–	33.0
GraphWalker-QA	[2]	–	–	–	30.2

Table 1: Exact match and latency.

4 Limitations

We did not evaluate on TriHop-Bench.

References

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